

Systematic Review

Expression profile of cancer stem cell markers SOX2, OCT4 & NANOG in salivary gland malignancies: A systematic review

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Received June 6, 2024; Accepted March 18, 2025; Ahead of print July 22, 2025; Published August 8, 2025

Background & objectives: Cancer stem cells influence aggressive biology, metastasis, recurrence, and treatment resistance in various malignancies. The transcription factors SRY-box transcription factor 2 (SOX2), Octamer-binding transcription factor 4 (OCT4), and Homeobox protein NANOG (NANOG) are prime controllers of the signalling circuit required for embryonic stem cell pluripotency. Salivary gland tumours exhibit diverse biological and clinical behaviours ranging from a benign, innocuous nature to highly aggressive tumours, with a great tendency for recurrence, and poor prognosis. Advances in therapeutic modalities have also been limited. This systematic review aims to uncover the differential expression and influence of SOX2, OCT4, and NANOG in salivary gland malignancies. This could help the stratification of high-risk patients and the identification of newer prognostic and predictive remedial targets.

Methods: PubMed, Scopus, Google Scholar, and Clinical key databases were searched for relevant articles, and studies that met the eligibility criteria were selected.

Results: Ten articles that fulfilled the eligibility criteria were included. All the studies supported the role of the studied markers as prognosticators and potential therapeutic targets.

Interpretation & conclusion: The aforementioned transcription factors might have contributed to aggressiveness and poor prognosis. Thus, it has been inferred that a combination of these factors may serve as a marker to determine the behaviour and therapeutic approaches for salivary gland malignancies.

Key words Cancer - pluripotency - prognosis - salivary gland - stem cells - therapeutic

Malignancies originating from salivary glands are diverse and bear complex pathologies exhibiting varied biological behaviour, clinical profiles, and histological and molecular diversities, resulting in considerable diagnostic, prognostic, and therapeutic

challenges¹. For patients' benefit and clinical and scientific research purposes, identifying and exploring biomarkers is highly relevant to improve survival rates and develop targeted therapies². The distinctive subpopulation of tumour cells having a primitive

state and enhanced invasive properties are cancer stem cells (CSCs), which are imperative for tumour development, metastasis, and relapse³. It has been suggested that the presence of the CSC population is involved in tumour initiation, progression, drug resistance, and recurrence. In the past few decades, the sustenance of CSCs in tumours has been substantiated; still, exploring markers that can identify these cells for possible therapeutic implications is ongoing⁴. Scientific research has focused on the immune-expression of stem cell markers and linked their immune profile with clinic-pathological characteristics of the tumours⁵. Cluster of differentiation 44 (CD44), octamer-binding transcription factor 4 (OCT4), and homeobox protein NANOG (NANOG), SRY-box transcription factor 2 (SOX2), aldehyde dehydrogenase (ALDH), B cell-specific moloney murine leukemia virus integration site 1 (Bmi-1), zinc finger protein SNAIL (Snail), Twist-related protein 1 (Twist1), keratin 14, CD133, mesenchymal-epithelial transition factor (c-Met), and CD24 are a few biomarkers implicated in the recognition of tumour stem cells in head and neck carcinomas and salivary gland malignancies⁶. A combination of CSCs has a plausible role in diagnosing and prognostication of salivary gland malignancies, but their exact pathobiology in individual salivary gland tumours is ambiguous⁷.

OCT4, SOX2, and NANOG are the transcriptional regulators that coordinate stemness characteristics extensively and are rich in both CSCs and embryonic stem cells (ESCs). The pluripotent circuitry is upheld within ESCs, and combined positivity for these three markers could reflect a high concentration of CSCs. In addition, these proteins regulate the expression and repression of differentiation and pluripotency-associated genes⁸. SOX2 is critical for pluripotency and is reported as a key player in oncogenesis. OCT4 is a crucial regulator of epithelial-mesenchymal transition (EMT), and NANOG is involved in blocking differentiation. These three factors form a closely knit circuit with other regulatory proteins and thus could be explored as potential targets to identify CSCs and develop CSC-targeted therapies⁹. Evidence-based data are needed to document the expression profiling of studied biomarkers in different salivary gland malignancies and their likely association with outcome so that future studies can be planned considering the strengths and limitations of the available literature.

Therefore, this systematic review was designed for a comprehensive and critical analysis of studies

that evaluate the role of SOX2, OCT4, and NANOG as diagnostic, prognostic, and therapeutic indicators in salivary gland malignancies.

Materials & Methods

The protocol of this systematic review was registered at the International Prospective Register of Systematic Reviews (PROSPERO) database (CRD42023408336) and conducted in agreement with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines.

Research question: This systematic review addressed the following research question: Do SOX2, OCT4, and NANOGSCMs determine the behaviour and prognosis of salivary gland malignancies? The Population Intervention Comparison Outcome (PICO) used to define the research question was:

Population: Patients with histologically confirmed salivary gland malignancy.

Intervention/Markers of interest: Immunohistochemical expression of CSC markers SOX2, NANOG, and OCT4 in salivary gland malignancies.

Control/Comparator: Normal salivary gland tissue/benign salivary gland tumours.

Main outcome: Assessment of immune expression of stem cell biomarkers SOX2, NANOG, and OCT4 to determine the predictive and prognostic efficacy in salivary gland malignancies.

Eligibility criteria: The eligibility criteria for study selection were determined before the literature search. Original research papers (randomised controlled trials, cohort, case-control, and cross-sectional) that evaluated the behaviour and prognostic and predictive effect of SOX2, OCT4, and NANOG immunohistochemically in salivary gland malignancies, published in the English language only, and which were freely available online with full-text version were incorporated.

Book chapters; meta-analysis; systematic reviews; review articles; case reports/case series; abstract-only articles; *in vitro* and animal studies; studies which had no relevance to the topic; conference abstracts; dissertations; duplicate articles; articles in a language other than English; articles with data variability; articles for which full text was not available online; studies which used techniques other than

immunohistochemistry to study the expression of stem cell markers; studies which evaluated CSC markers other than SOX2, OCT4, and NANOG were excluded.

Search strategy: Articles were identified through an electronic search in four databases (PubMed, Scopus, Clinical Key, and Google Scholar) from April 16 to April 30, 2023 without any period restriction. The keywords included were ‘Cancer,’ ‘SCMs,’ ‘SOX2,’ ‘OCT4,’ ‘NANOG,’ and ‘Salivary gland malignancy’ with the use of Boolean operators “OR” and “AND” between them. A bibliographic list of included articles was also checked for any additional studies meeting the eligibility norms.

Article screening and eligibility evaluation: Two reviewers (DS and SG) independently screened the titles and abstracts of all the articles, and those not fulfilling the eligibility criteria were excluded. The same two authors then read the full text of the articles to evaluate eligibility. The differences of opinion were sorted by discussion, and consensus was achieved by consulting a third reviewer (MK).

Data extraction: To certify the reliability of the content, data were extracted by the first author (DS) and revised by the second author (SG). The data were extracted from the included studies: general information (authors; article title; year of publication); study characteristics (objectives; study design; inclusion and exclusion criteria, study inferences); participant characteristics (number of participants enrolled, age, gender, clinicopathological disease characteristics); immunohistochemical expression of CSCs by SOX2, OCT4, and NANOG in salivary gland malignancies; outcomes. A narrative synthesis structured around the study population characteristics and outcome measures was performed to extrapolate the possible prognostic or predictive significance of SOX2, OCT4, and NANOG derived from the findings of the included studies (Table I and II)¹⁰⁻¹⁹.

Quality assessment: The two reviewers (SG and DS) independently assessed the methodological quality of the included articles, and contradictions were sorted after consultation with a third author (MK). The Newcastle-Ottawa Scale (NOS) was used to assess the quality of the included articles for case-control/cross-sectional studies, and articles with seven or more stars were regarded as high quality.

Results

Study selection: Out of the 218 articles identified from the databases, 126 duplicate studies were removed. Further, 92 articles were screened, and 31 were assessed for eligibility. After considering the desired inclusion criteria and identifying articles from other sources, 10 studies that met the eligibility criteria were included (Fig. 1).

Quality assessment: All the studies were of high quality, with scores 7 and 9 in six and four studies, respectively (Fig. 2 and 3).

Study characteristics: Among the 10 included studies, five were case-control, and five were cross-sectional in design. The total number of cases documented in the included studies was 541, of which 264 were males and 245 were females, respectively. Adenoid cystic carcinoma (ACC)^{10,14,16} and mucoepidermoid carcinoma (MEC)^{11,13} were found in three and two studies, respectively, and one each was associated with adenocarcinoma¹⁹ and squamous cell carcinoma of salivary glands¹⁷. The remaining three^{12,15,18} documented a combination of salivary gland malignancies. Tumour characteristics like histological grading and staging were documented in six^{1,10,12-14,16,19} studies, histological grading only in three cases^{11,17,18}, and none reported clinical staging alone. Four studies included tumours of both^{11,12,16,18} minor and major salivary glands, while three^{14,17,19} and one¹³ study evaluated major and minor salivary gland malignancies, respectively. The follow up period was documented in five^{10,12,13,17,19} and recurrence was mentioned in three studies^{12,17,19}. Expressions of SOX 2 and OCT-4 were individually assessed in six^{10,12,14,15-17} and one¹⁸ research articles, respectively, while two studies evaluated a combination of SOX2 and NANOG^{13,19} and only one study assessed OCT4 and NANOG¹¹. None of the studies evaluated the individual expression of NANOG. All the studies supported the role of the studied markers as prognosticators and potential therapeutic targets.

Discussion

Cancer seeding and advancement are governed by a small subpopulation of self-renewing and differentiating oncogenic CSCs²⁰. In salivary gland malignancies, characterisation of CSCs could provide a better insight into the pathobiology and aid in

Table I. Clinicopathological disease characteristics of the participants involved in different studies

S. No.	Study Reference	Study Design	No. of cases/control	Age (yr) Gender (M:F)	Type of Neoplasia	Site	TNM	Histological grading	Origin	PNI Y:N	Metastasis Y:N	Follow up	Recurrence
1.	Dai <i>et al</i> ¹⁰ , 2014	Case-Control	Cases -131 Control-131 (paired Adjacent non-cancerous tissue)	36-78 75:56	ACC		I or II: 77 III or IV: 54 (UICC 2002 staging system)	Cribriform/ Tubular-102 Solid-29 (WHO classification)	-	95:36	43:88	3 - 62 months	-
2.	Rodrigues <i>et al</i> ¹¹ , 2016	Case-Control	Cases -28 Control-20 (Non-neoplastic salivary gland tissue from same patient)	45.5 30% (8):70% (20)	MEC	Ma/Mi-19/9 PA-16 SM-03 Palate-04	PA- T1/T2-10 T3/T4-15 CXP	Low/Int/High-21:4:2 (WHO grading)	-	9:19	-	--	-
3.	Sedassari <i>et al</i> ¹² , 2017	Case-Control	Cases-25 Control-30 (CXP) Control-30 (PA)	CXPA/PA 61.4/38.5 12:13/19:11	PA-30 CXP-25 Intra/ Extracapsular -7:18	PA- P/SM/Mi-15/1/14 CXP- P/Mi-16/9	PA- T1/T2-10 T3/T4-15 CXP	CXPA- High/Low-19/6 (WHO grading)	Primary -19 Recurrent -06	Intra/Extra- capsular-0/7	Intra/Extra- capsular-0/10	2000 days	Intra/Extra- capsular-0/11
4.	Xu <i>et al</i> ¹³ , 2017	Cross-sectional	75	42:49 25:50	MEC of palate	HP-74.7% Jxn-22.7% SP-2.6	T1 & 2-90.7% T3 & 4-9.3% (AJCC classification)	Low/Int/High-38/31/06 (Brandie's classification)	-	0/75	-	11 to 124 months	-
5.	Thierauf <i>et al</i> ¹⁴ , 2018	Cross-sectional	45	55 18:27	ACC	Parotid-15 Extra- parotid-30	I,II-23 III,IV-22 (AJCC classification)	Cribriform/tubular/ mixed-32 Solid-13	-	-	2/43	-	-
6.	Kamal <i>et al</i> ¹⁵ , 2018	Cross-sectional	32	-	MEC -16 (low grade-high grade/8:8) PSSC-16 (low grade: high grade/8:8)	-	-	-	-	-	-	-	-
7.	Luo <i>et al</i> ¹⁶ , 2019	Case-Control	Cases-34 Control- 10 (Paracancerous normal gland tissue)	30-78 24:20	ACC	P/SM/ SL 18/5/15 Palatine-6	Stage I/II:III/IV 21:13	Sieve/Tubular: Solid=26:8	-	-	-	26-66 months	-
8.	Bertlha <i>et al</i> ¹⁷ , 2021	Cross-sectional	31	77 (42 -92) 19:12	SCC	P/SM 27/4	-	G1-2 G2-26 G3-03	Primary -18 Metastatic-13	-	-	-	16
9.	Mourat <i>et al</i> ¹⁸ , 2021	Case-Control	Cases-60 Control -05 (mucus extravasation phenomenon and remnant parenchyma of a normal salivary gland for the Ma/MiSGT)	44.7 35:25	PA/MEC/ACC-20 each Ma/Mi 10:10 for every tumour	P/SM/ SL 18/10/2 Palate -14 Lip mucosa -6	-	PA- Cellular/Classical/ Myxoid:9/6/5 (Seifert <i>et al</i>) MEC- Low/Int/High: 10/3/7 ACC- Solid/Cribriform/ Tubular:10/6/4 (Barnes <i>et al</i>)	-	-	-	-	-
10.	Spiegel <i>et al</i> ¹⁹ , 2021	Cross-sectional	40	64.4 29:11	Adenocarcinoma of the salivary glands	P/SM-28/12	T1-2 /T3-4 -20/20 (UICC 7th edition)	G1/G2/G3 -8/21/11 08/21/11	-	14:26	-	41.8 ± 42.0 months	10

P, parotid gland; SM, submandibular gland; SL, sublingual; PNI, perineural invasion; PA, pleomorphic adenoma; PSSC, primary squamous cell carcinoma; SCC, squamous cell carcinoma; ACC, adenoid cystic carcinoma; MEC, mucoepidermoid carcinoma; CXP, carcinoma ex-pleomorphic adenoma; SGT, salivary gland tumour; Mi, minor; Ma, major; Int, intermediate; SP, soft palate; HP, hard palate; Jxn, junction; AJCC, American Joint Committee on Cancer; WHO, World Health Organization; UICC, Union for International Cancer control

Table II. Expression profile of various Immunohistochemical markers and the interpretation

S. No.	Study	Markers Used	Expression evaluation	Immunoeexpression & findings	Inference	Limitations
1.	Dai <i>et al</i> ¹⁰ , 2014	SOX-2	Total score was calculated as: Staining score = intensity × positive rate (cell staining intensity and percent of positive cells)	-High:Low/82:49 -High SOX2 expression patients had lower OS and DFS (5yr) -High:Low- (SOX2 expression)- Hazard Ratio: 4.71: 4.55. -Adjacent normal tissue was negative for SOX2 expression.	-A significant biomarker to evaluate the ACC patient's prognosis may be SOX2. -Elevated expression of SOX2 was significantly associated with T category and distant metastasis. -SOX2 expression and age, gender, histological type, and nerve invasion showed no significant association.	Not mentioned
2.	Rodrigues <i>et al</i> ¹¹ , 2016	OCT-4 NANOG	Positive tumour cells percentage was evaluated and categorised as negative (0-10%), 1 (10- 50%) and 2 (> 50%)	-OCT-4 Negative/Score 1/Score 2: 8/7/13 NANOG Negative/Score 1/Score 2: 7/8/13 -2 non-cancerous salivary gland tissue revealed OCT-4 expression in the cytoplasm of ductal cells. In addition, discrete number of acinar cells also showed positive expression. -Positive NANOG expression was seen in ten non-cancerous gland tissue. -Statistically significant relationship was observed between OCT-4 expression with PNI, growth pattern, desmoplasia, margin status. -Statistically significant relationship was observed between expression of NANOG with PNI, desmoplasia.	Expression of OCT4 and NANOG contribute to worst prognosis and aggressiveness	Non-inclusion of correlation analysis between the immunoeexpression of CSC markers with overall survival
3.	Sedassari <i>et al</i> ¹² , 2017	SOX-2	At least 5% of nuclear stain positivity of neoplastic cells was considered positive. Further subcategorised as focally (>5-50%) and diffusely positive (>50%) cells.	Total number of positive cases PA -0 CXPA- 24 Positivity for SOX2 Intracapsular-85.7% Extracapsular - 100%	-Diffuse SOX-2 expression had a significant correlation with high histological grade, advanced pT stage, recurrence and distant metastasis. -Patients had shorter OS rate with extracapsular-CXPA and diffuse SOX2 expression -No significant relation between SOX-2 expression and DFS rate. -SOX2 is significant for carcinogenesis and tumour invasion in CXPA	Less number of cases, geographic and therapeutic differences, retrospective use of paraffin-embedded tissues may impact tissue antigenicity

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S. No.	Study	Markers Used	Expression evaluation	Immunoeexpression & findings	Inference	Limitations
4.	Xu <i>et al</i> ¹³ , 2017	NANOG SOX2	Immunopositive cancer cells percentage was evaluated and graded as strongly positive (>50%) weakly positive (10–50%) and negative (0–10%),	-Negative/Weakly, Positive/Strongly positive -SOX-2-29/27/19 -NANOG-23/32/20 -No significant association of SOX-2 and NANOG with 5-yr OS rate.	-SOX-2 and NANOG alone might not be effective prognosticators for MEC of palate.	Incomplete information collection due to missing data for patients with distant metastasis or regional recurrence
5.	Thierauf <i>et al</i> ¹⁴ , 2018	SOX2	Immunoeexpression was evaluated as (1=0%, 2= less than 33%, 3= between 34% and 66%, 4= more than 66% positive tumour cells).	Prominent nuclear staining for SOX2 was observed in 31% of ACC.	-Absence of SOX2 expression seen in solid ACC, characterised by more aggressive phenotype. -No significant correlation of SOX2 expression profile was observed with clinical data. Higher SOX2 expression in patients with a lower tumour stage and without lymph node metastasis. A significantly high SOX2 expression was seen in high-grade tumours.	-Detailed information of different subgroups was lacking -Small size cohort
6.	Kamal <i>et al</i> ¹⁵ , 2018	SOX2	Immunostaining was considered as SOX2 positive reaction with visible brown coloured nuclear stain. The area percentage of immune positive cells to the area examined using computerised image analyser was estimated to assess immunoreactivity	-MEC and PSSC revealed positive nuclear expression for SOX2. -A statistically significant difference and higher mean value of SOX2 area percent of immunoeexpression was documented in PSSC high grade, while followed by MEC high grade, while the lowest value was recorded in PSSC low grade.		Not mentioned
7.	Luo <i>et al</i> ¹⁶ , 2019	SOX-2		Positive expression of SOX-2 (IHC)--- Cases (47.06%) SOX-2 Positive expression- T1/T2 :T3/T4-28.57%:76.92% Sieve/Tubular: Solid- 34.6%:87.5% One of the 10 cases in the control group showed low expression.	-SOX2 expression correlated with tumour cancer stem cells -High SOX2 expression in the periphery of the tumour suggested it's relation with invasion and aggressiveness of tumour.	Not mentioned
8.	Berticha <i>et al</i> ¹⁷ , 2021	SOX-2	Immunoreactivity was quantified with H-score. The formula used was 3 × percentage of strongly staining membrane/ cytoplasm/nuclei + 2 × percentage of moderately staining membrane/ cytoplasm/nuclei + percentage of weakly staining membrane/ cytoplasm/nuclei, giving a range of 0–300. ROC curve and Youden Index were used to derive cutoff value to define high or low marker expression.	high:low/no of cases- 15:16 high:low OS (%):37.1:62.3 (3yr); 37.1: 49.9 (5 yr) DSS (%):59.7: 80 (3 yr); 59.7: 80 (5 yr) RFS (%):21.9: 62.5 (3 yr); 21.9: 62.5 (5 yr) LCR (%):48.2: 75.0 (3 yr); 48.2: 75.0 (5 yr)	-SOX-2 expression correlated with poorer prognosis. -High expression of SOX-2 indicates increased CSC activity and require more aggressive therapy. -Increased RFS rates associated with low expression of SOX-2.	Not mentioned

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S. No.	Study	Markers Used	Expression evaluation	Immunoeexpression & findings	Inference	Limitations
9.	Moura <i>et al</i> ¹⁸ , 2021	OCT-4	The scores of percentages of positive cells (PP) and intensity of expression (IE) were summed to obtain the total score, that ranged from 0 to 7.	-With regard to tumour type and anatomical location most frequent percentage of positive cells and the intensity of expression score for OCT-4 was 4 (n = 40) and 3 (n = 30) -Total immunostaining score was Ma/Mi SGT-6.6/5.3 Benign/Malignant- 5.4/6.2 -Comparison of OCT4 and CD44 expression between MaSG and MiSG tumours revealed lower expression in the latter.	-Biological behaviour of salivary gland neoplasms is indicated by immunoeexpression of tumour stem cell markers -OCT4 proteins were significant markers to identify tumour stem cells. -Both luminal and non-luminal cells were positive for OCT-4 in the case of PA and ACC. It was positively expressed in squamous and intermediate cells but not in mucous cells in MECs thus highlighting a less aggressive phenotype.	Not mentioned
10.	Spiegel <i>et al</i> ¹⁹ , 2021	NANOG SOX2	H-score was used to quantify immunoreactivity. -Youden Index were used to derive cutoff value to define high or low marker expression.	NANOG- Low:High 36:4 SOX2- Low:High 33:7 NANOG OS%- high:low—66.7/71.6 (3 years); -0: 66.1 (5yr) DSS%- high:low—66.7/84.2 (3 years); 0: 84.2 (5 yr) RFS%- high:low—50/74.5 (3 years); 0:74.5 (5 yr) LCR- high:low—75/85.5 (3 yr) ;0: 85.5 (5 yr) SOX2- OS%- high:low—100/64.1s (3 yr) ; 75/64.1% (5 yr) DFS %- high:low—100/77.6 (3 yr) ; 100/77.6 (5yr) RFS%- high:low—71.4/72.5 (3 yr) ; 71.4/72.5 (5 yr) LCR%- high:low 83.3/84.8 (3 yr) ; 83.3/84.8 (5 yr)	-Significant association of high expression of SOX-2 with negative PNI was observed. -Low expression of NANOG/SOX-2 was associated with higher grading. - Survival-estimates had no significant correlation with investigated CSCs markers.	-Heterogenous methods applied, cohort and Retrospective analysis

DFS, disease free survival; DSS, disease-specific survival; G, grading; LCR, local control rate; OS, overall survival; RFS, recurrence-free survival; T, tumour size; N, nodal category; M, metastasis; pT, pathological tumour size

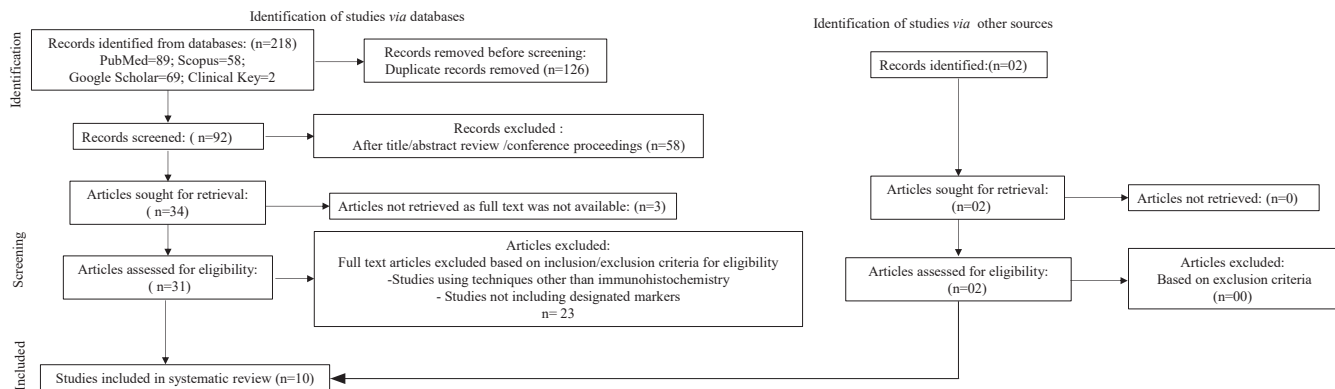


Fig. 1. Flowchart depicting Identification, screening and inclusion of the article.

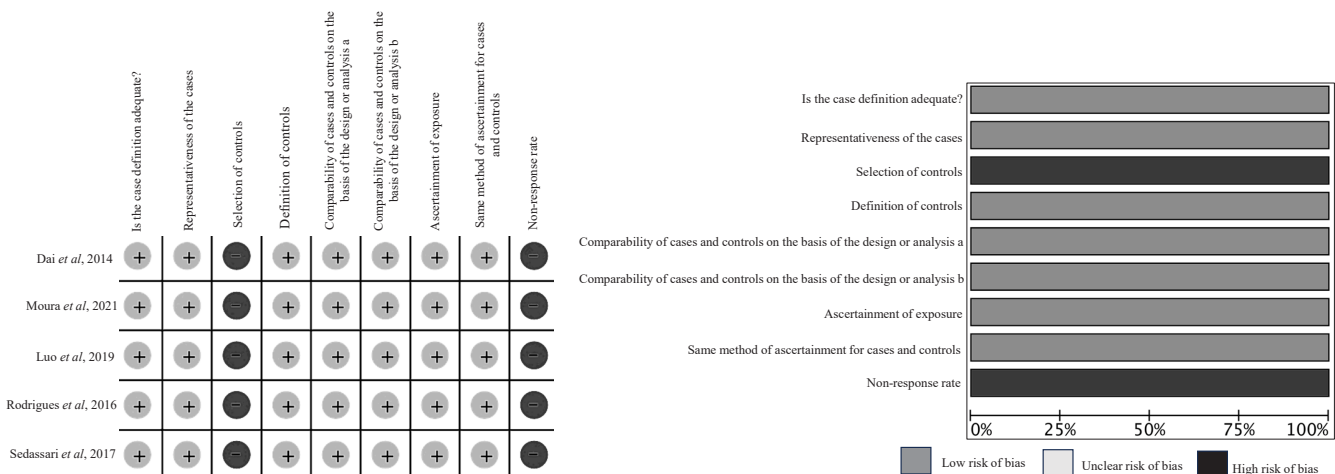


Fig. 2. Risk of bias assessed by the Newcastle-Ottawa quality assessment scale for case-control studies.

the development of more effective therapies. Thus, currently, squamous cell metaplasias (SCMs) are being investigated to identify CSCs, and their expression profile is correlated with the clinicopathological features²¹. Molecular and immunohistochemical markers are a boon for CSC research as they help to identify and isolate CSCs and explore the pathways of oncogenesis, thus, establishing a novel target for therapeutic purposes¹¹.

The core transcription factors SOX-2, OCT4, and NANOG play an imperative role in the retention of the pluripotent ESC phenotype through self-controlling, feedback, and feed forward regulatory circuits²². These ESC biomarkers are reported to have a strong correlation with CSCs as both share specific tumour characteristics, and certain mutual molecular cross-talks might exist between the two celltypes¹⁹. Biomarkers SOX-2, OCT4, and NANOG independently and synergistically interact bringing transcriptional alteration in the genetic pathways. SOX2 and OCT4 have a common

DNA-binding site, thus regulating the target genes by forming heterodimers. In addition, SOX2 keeps OCT4 expression at suitable levels²³. Moreover, NANOG is an established target modulated by OCT4/SOX2; thus, a well-linked transcriptional control looped pathway has been recognised²⁴. Furthermore, these proteins have been substantially related to EMT and various clinicopathological factors². The early transcription factors can mediate tumour initiation, proliferation, differentiation, metastasis, relapse, and therefore, increased expression of studied proteins has been linked with worse prognosis and dismal outcomes in various human malignancies²⁵. The elucidation of the molecular intricacies causing varied expression of CSC markers in salivary gland malignancies through additional clinical or experimental studies might help to substantiate these transcription factors as unique potential targets for predictive and therapeutic approaches²⁶.

In salivary glands, the intercalated and excretory ducts are the locus for proliferating cells, and acinar

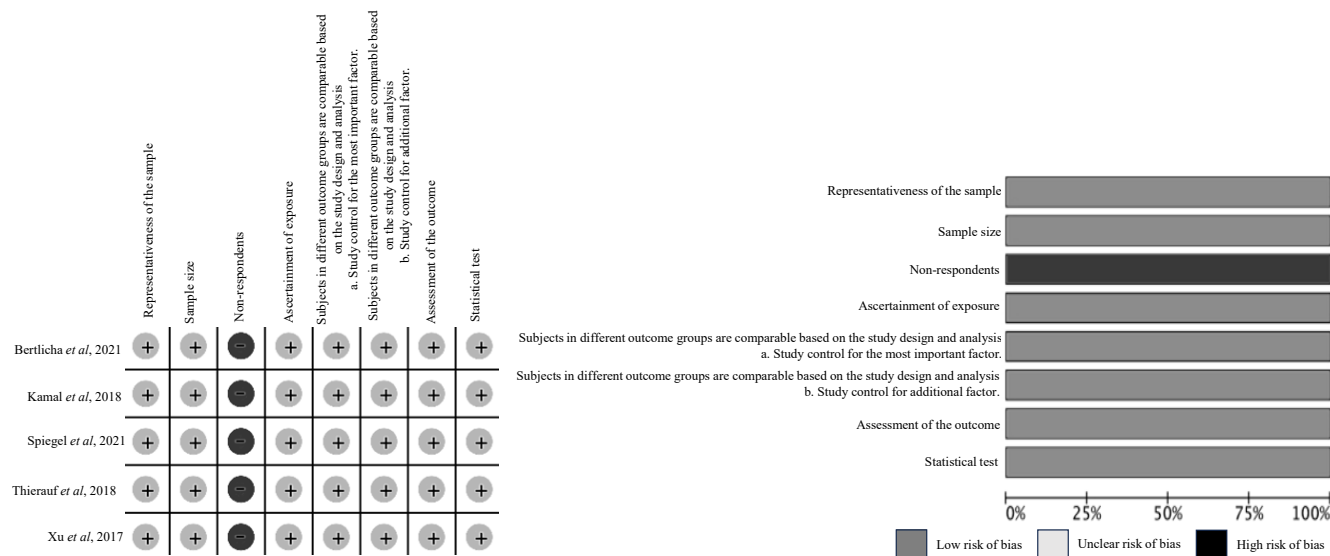


Fig. 3. Risk of bias assessed by the modified Newcastle-Ottawa quality assessment scale for a cross-sectional study.

components have a limited degree of proliferative ability, implying that a population of potent progenitor stem cells resides in the ducts¹⁰ SOX-2 protects CSCs against apoptosis and is significantly responsible for their longevity. Highly diverse expression patterns and correlation with clinicopathological outcome are suggestive of the distinctive role of SOX2 in individual tumours²⁷. In salivary gland carcinomas, higher SOX2 expression indicates increased CSC activity and, therefore, could be a reliable marker for predicting and may be a potential therapeutic target requiring more aggressive therapy²¹. Dai *et al*¹⁰ in ACC observed poorer overall and disease-free survival with higher expression of SOX2. Higher expression of CD44 and SOX2 in primary squamous cell carcinoma and high-grade MEC suggests that therapies combining these markers might benefit the patients¹⁶. In addition, Xu *et al*¹³ stated that the amalgamation of biomarkers SOX2/CD44/CD133 has prognostic significance in the MEC of the palate. However, Sadeghi *et al*²⁸ found a considerable link between higher grades of MEC and low SOX2 expression and poor outcomes in ACC patients. Luo *et al*¹⁶ reported significantly marked expression of SOX2 and surviving in ACC as compared to normal salivary gland tissues and stated that both markers are tissue-specific and may become salivary gland ACC oncogenes. They also observed increased SOX2 expression in the peripheral region of the tumour pointing towards invasive, aggressive, and malignant behaviour of the tumour. Bertlacha *et al*¹⁷ also found that elevated SOX-2 expression is associated with disease

aggressiveness and a poorer prognosis. Thierauf *et al*¹⁴ postulated a context-dependent function and regulation of SOX2 in distinct malignancies.

Human adult stem cells, germ cells and ESCs express transcription factors OCT4 and NANOG with higher proliferative capacity leading to aggressive tumour biology²⁹. The potential of NANOG and OCT-4 to act as a prognostic biomarker has been explored by researchers in the malignancies of salivary glands and other head and neck regions³⁰. An association between the NANOG and OCT4 expression with perineural invasion, other histopathological features, and positive expression in ductal structures in MEC has been related to destructive behaviours and worst outcomes. In addition, tumour-initiating abilities and metastasis have been controlled by OCT4/NANOG signalling through EMT¹⁸. Higher OCT-4 expression in ACCs and MECs has been attributed to characteristics inherent to embryonic development through EMT²⁹. However, Spiegel *et al*¹⁹ observed lower NANOG and SOX2 expression without any noteworthy association with survival estimates and stated that both markers have no prognostic value in salivary gland adenocarcinoma. Santos *et al*³¹ found raised expression of OCT-4 and SOX2 in lesions without myoepithelial differentiation and malignant behaviour. However, stromal immunoexpression of OCT4 was higher in malignant salivary gland tumours and was related to myoepithelial differentiation.

The present systematic review provides evidence for the higher expression of pluripotent CSC markers in malignant salivary gland tumours and their significant association with higher-grade and advanced-stage malignancies, although contradictory findings about therapeutic effectiveness were present. Therapeutic modalities of salivary gland malignancies are particularly challenging because of disease rarity and heterogeneity¹. Poor prognosis and dismal median overall survival (OS) of 15 months and five-year OS rates of about 15 per cent have been seen in patients with recurrent or metastatic (R/M) disease³². Therefore, selectively targeting CSCs might hinder recurrence and metastasis of the tumour thereby improving outcome and patient's quality of life¹⁶. Experimental research has documented that induced pluripotent stem cells (iPS) could treat salivary gland tumours and restore glandular function³³. The rationalisation of therapeutic strategies tailored to the patient's specific cancer type and prognosis of R/M disease can be significantly improved by unravelling immunohistochemical expression profiles and mapping genomic variations in salivary gland malignancies, thus justifying and warranting biomarker-driven research^{34,35}.

There were few limitations in this systematic review. The included studies in the present systematic review are retrospective and have used immunohistochemistry as a primary investigative tool, and paraffin-embedded tissues archived for longer periods could affect immunohistochemical expression of the markers. Most of the studies have smaller sample sizes because of the low incidence of certain salivary gland malignancies and the non-availability of clinical prospective trials. Studies used different therapeutic approaches that could be one of the confounding factors for clinical outcomes. The inability to conduct meta-analysis is one of the major limitations of the present systematic review because of the lack of relevant data about the expression profile of CSC markers with survival rates, grading, and clinical stages. In addition, the number of studies retrieved on specific salivary gland malignancies was less, making meta-analysis impossible. Another major constraint was the inability to explore heterogeneity due to a lack of relevant data. In addition, no access to some full-text articles, non-inclusion of studies in a language other than English, and some that failed to produce salivary gland malignancy-specific data could also result in publication bias.

Overall, in malignancies, substituting conventional treatments with more efficient options have been revolutionised by the emergence and detection of CSCs by specific biomarkers with a deep influence on the expansion of molecular therapeutic strategies. SOX2, OCT-4, and NANOG have been proven as prognosticators in salivary gland malignancies, with a definite role in tumour biology. The selective targeting and unravelling of tumorigenic CSCs, genetic alterations, and protein expression profiles of salivary gland malignancies could rationalise treatment modalities tailored to the patient's tumour.

Financial support & sponsorship: None.

Conflicts of Interest: None.

Use of Artificial Intelligence (AI)-Assisted Technology for manuscript preparation: The authors confirm that there was no use of AI-assisted technology for assisting in the writing of the manuscript and no images were manipulated using AI.

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